Eshita Kundu

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Summary

Applied AI engineer building LLM agents, MCP servers, and tool-driven workflows, from FastAPI backends to automated evals and cloud deployment. Focused on reliable agent execution: bounded tools, validated outputs, and deterministic fallbacks.

Technical Skills

Languages & frameworks: Python, TypeScript, SQL, FastAPI, Pydantic, SQLAlchemy, React
Agents & LLMs: MCP / FastMCP, LangGraph, LangChain, LlamaIndex, OpenAI & Anthropic SDKs, NVIDIA NOOA, n8n
Models & retrieval: OpenRouter, NVIDIA NIM, Hugging Face, Ollama, pgvector, Chroma
Evaluation & testing: LangSmith, Langfuse, Ragas, Promptfoo, pytest, Playwright
Data & infrastructure: PostgreSQL, MongoDB, SQLite, Docker, GitHub Actions, AWS, Cloudflare, Vercel, DigitalOcean

Projects

TraceIntel - On-Chain Permission Risk Agent Live · Demo · GitHub **2026**
Python, FastAPI, NVIDIA NOOA, OpenRouter, PostgreSQL, React, Docker, GitHub Actions

- Engineered a **bounded LLM investigation agent** with 3 read-only tools, a fixed call budget, and **deterministic evidence validation**; invalid output gets one repair attempt, then falls back to rule-based analysis.
- Built the pipeline that decodes ERC-20 approvals from **Ethereum and Monad** transactions and compares each permission's original and current on-chain state (allowance, balance, contract code).
- Wrote **24 offline agent eval scenarios** that run in CI, including **5 adversarial cases** (prompt injection, out-of-scope tool calls, fabricated claims), separate from ~95 unit and browser tests.
- Implemented per-client rate limiting, daily LLM cost controls, and persistent agent execution traces for debugging.

Agon - Guardrailed Agentic Trading on Solana (team) Site · Demo **2026**
TypeScript, MCP, Solana, Jupiter, Swig smart wallets

- Team-built **agent-agnostic MCP trading server**: AI clients paper-trade Solana tokens at live Jupiter prices, and a **deterministic guard outside the model** checks every trade before signing.
- Designed the **agent-safety model**: a vault-enforced slippage floor and per-trade spend cap, and a signing rule that executes only guard-cleared trades, never parameters the agent wrote.
- Defined **fail-closed recovery** for unknown trade outcomes: no blind retries, idempotent trade IDs chosen by the agent, and full rule disclosure before an owner arms it.

ARIS - Software Adoption Research Workflow Live · Demo · GitHub **2026**
Python, Heym, NVIDIA NIM, Tavily (MCP), Docker, DigitalOcean, Cloudflare

- Designed a **hybrid AI workflow**: 7 parallel research branches gather evidence (GitHub, OSV advisories, PyPI/npm, Stack Exchange, web search), LLMs extract structured findings, and the result is emailed as an adopt / trial / hold / avoid brief.
- Separated LLM research from **deterministic decision-making**, using Python to compute weighted scores, apply security vetoes, and calculate confidence.
- Built a Python execution node for the Heym workflow engine (7 tests, proposed upstream) and self-hosted it with Docker.

Study Buddy - MCP Server for Studying GitHub **2026**
Python, MCP SDK (FastMCP), SQLite, Pydantic

- Built an MCP server for Claude Desktop with **13 tools, 3 resources, and 4 prompts** over local notes, PDFs, and past papers, with SQLite mastery tracking and path-traversal protection.

More projects: remote FastMCP ranking server, multi-agent research workflow, Telegram DevOps agent (team) | full list

Experience

Employability.life | RPA Developer & Business Analyst **Sep 2024 - Nov 2024**
Experiential learning program with Federation University Australia *Kolkata, India*

- Automated manual Excel reporting with UiPath, saving **10+ hours per week** and reducing errors by **70%**; wrote process design documents and led requirements, testing, and sign-off across Agile sprints (certificate, recommendation).

Education

Sister Nivedita University **Jun 2022 - Jun 2026**
Bachelor of Technology in Computer Science and Engineering; CGPA: **8.80/10**

Achievements

- **2nd Place, BRICS-FS-36 Data Analysis & Visualization**, international final, China, Dec 2024 (certificate)
- **EL Excellence Award, Top 50 Learner**, Federation University Australia, Feb 2026 (award, citation)

Certifications

MCP Mastery, Fractal Analytics / Coursera | **Applied Machine Learning in Python**, University of Michigan